

Challenges in Visual-Inertial SLAM



Visual drift in low-light

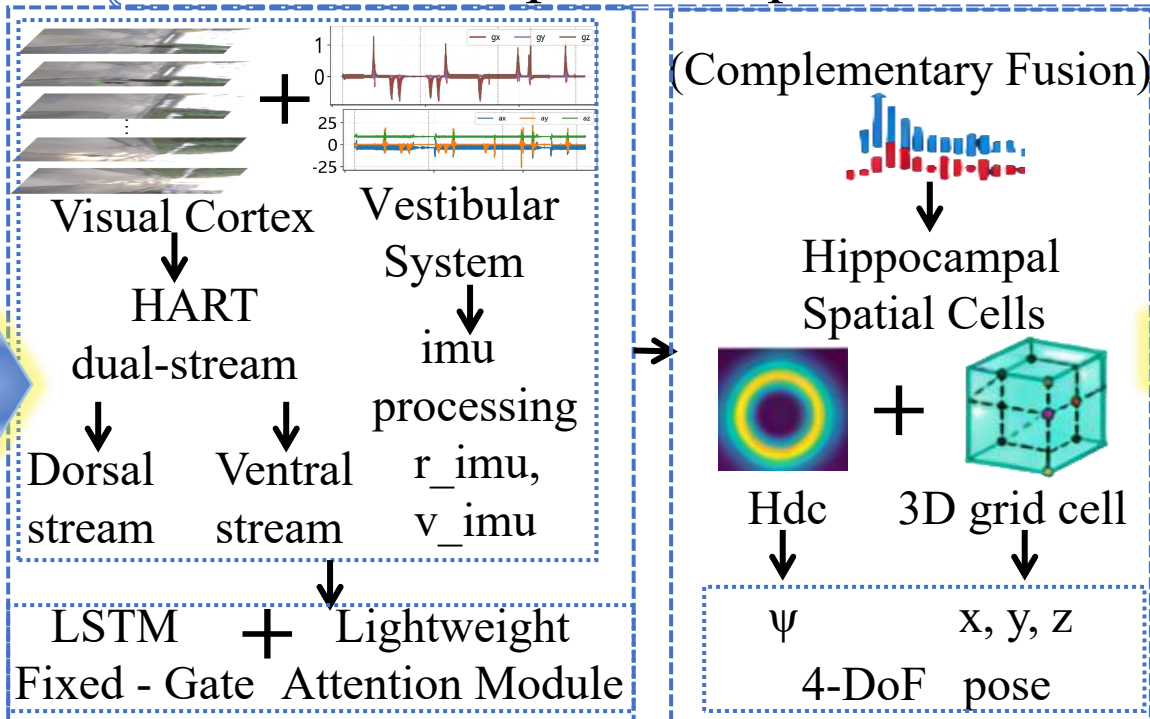


IMU bias accumulation



Loop closure failure

NeuroLocMap: Bio-Inspired Fusion



Robust Localization & Mapping



- 38.1% RMSE ↓
- 195 VT
- Real-time: ~31 FPS
- Zero-shot deployment

- ☒ Urban environments
- ☒ Indoor spaces
- ☒ Challenging lighting